

SAT Math Mastery · Week 2A

Linear Systems

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	number of solutions · the “second equation” · target expressions without solving · solutions and constants · parametric common points · graphs of systems · elimination shortcuts · category, mixture and rate systems
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Decide whether a 2×2 linear system has one, none or infinitely many solutions from its coefficients.
- S2.** Write or choose a second equation that forces a required number of solutions.
- S3.** Find a target expression ($x + y$, $2x - y$, $3x + 3y \dots$) by adding or subtracting equations, without solving for x and y .
- S4.** Use a given solution or intersection point to find unknown constants in the equations.
- S5.** Find the point shared by a whole family of lines ℓ_t by making the coefficient of the parameter vanish.
- S6.** Read a system from a graph (slopes, intercepts, intersection) and use it with a parameter.
- S7.** Choose the fastest elimination: add, subtract, scale, or use symmetry in the coefficients.
- S8.** Model counts, prices, mixtures, ratios and rate–time situations as a system and answer the quantity asked.

2 · REQUIRED FACTS

- The solution of a system is the ordered pair that makes *both* equations true; on a graph it is the intersection point.
- Two lines with different slopes meet exactly once; equal slopes with different intercepts never meet; equal slopes with equal intercepts are the same line.
- A system has infinitely many solutions only if one equation is a nonzero multiple of the other (every part, including the constant).
- Scaling one equation changes nothing; scaling only the left side and changing the constant creates a parallel line.
- A line $y = mx + b$ written as $Ax + By = C$ has $m = -\frac{A}{B}$; compare slopes before comparing constants.
- A problem often asks for an expression such as $x + y$ or a single variable: read the question before solving.
- Time = $\frac{\text{distance}}{\text{rate}}$; convert minutes to hours (or hours to minutes) before substituting.

3 · REQUIRED FORMULAS

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad \text{one solution: } \frac{a_1}{a_2} \neq \frac{b_1}{b_2}; \quad \text{none: } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}; \quad \text{infinite: } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\text{Cramer's rule (check only): } x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}, \quad y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$$

$$\text{Parametric family: } t(\text{expression}_1) + (\text{expression}_2) = 0 \Rightarrow \text{expression}_1 = 0, \text{ expression}_2 = 0 \quad d = rt, \quad t = \frac{d}{r}$$

4 · MINI THEORY SHEET

Elimination vs. substitution.

Look for	Move
a variable already isolated ($y = 2x - 1$)	substitute into the other equation
equal or opposite coefficients	add or subtract the equations
coefficients that are multiples of each other	scale one equation, then add or subtract
a question about $x + y$, $x - y$, $2x + 2y$	combine the equations to produce that expression directly
a graph with two lines	read two lattice points on each line, rebuild the equations, then solve

Number of solutions in one line. Compare $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$, $\frac{c_1}{c_2}$. Cross-multiplying is safer than dividing: $a_1b_2 = a_2b_1$ means parallel.

The second equation. For no solution, keep the left side (or a multiple of it) and use a constant that is *not* the matching multiple. For infinitely many, multiply the whole equation. For exactly one, any equation whose coefficient ratio differs.

Parameter conditions. Turn the requirement (no solution, infinite solutions, given point) into an equation in the constant; then check the constants too, because a value that makes the lines parallel may also make them identical.

Word systems. Name the unknowns in the units the question uses. A count equation and a value (or rate) equation are the usual pair; for ratio chains, express every count with one variable.

Speed shortcuts.

Situation	Shortcut
$x + y$ wanted, coefficients swap ($4x + y$, $x + 4y$)	add: $5(x + y) = \text{sum}$
$x - y$ wanted, same swap	subtract the equations
target is $px + qy$	find u, v with $u(\text{eq}_1) + v(\text{eq}_2) = px + qy$ (match both coefficients)
solution (x_0, y_0) given, constants unknown	substitute, then add/subtract the two new equations
parametric family	write $t(\dots) + (\dots) = 0$ and solve the two brackets
parallel test	$a_1b_2 = a_2b_1$; then compare the constants
mixture of two prices	total value – (all cheaper item) = (price gap) $\times$ (count of the dearer item)
two speeds, one total time	use $t = \frac{d}{r}$ for each leg, then add the times

5 · GUIDED EXAMPLES

Example 1 — Number of solutions and the second equation

The first equation is $4x - 6y = 9$. Give a second equation so that the system has (a) no solution, (b) infinitely many solutions, (c) exactly one solution.

(a) Same left side (or a multiple), wrong constant: $2x - 3y = 5$ (multiplying by 2 gives $4x - 6y = 10 \neq 9$). (b) Multiply *everything*: $8x - 12y = 18$. (c) Different coefficient ratio: $3x + y = 2$, since $4 \cdot 1 \neq (-6) \cdot 3$.

Example 2 — Target expression without solving

If $5x + 3y = 21$ and $3x + 5y = 19$, find $x + y$ and $x - y$.

Add: $8x + 8y = 40 \Rightarrow x + y = 5$. Subtract: $2x - 2y = 2 \Rightarrow x - y = 1$. (Check: $x = 3$, $y = 2$ works in both equations.)

Example 3 — Common point of a family of lines

For every real t the line $(t + 1)x + (2 - t)y = 3t + 6$ passes through one fixed point. Find it.

Collect the t terms: $t(x - y - 3) + (x + 2y - 6) = 0$. This holds for every t only if $x - y - 3 = 0$ and $x + 2y - 6 = 0$.

Subtract: $3y = 3 \Rightarrow y = 1$, so $x = 4$. The point is $(4, 1)$.

6 · PRACTICE SET — 36 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

What is the value of x in the system of equations below?

$$y = 2x - 1$$

$$3x + y = 19$$

(A) 3

(B) 4

(C) 7

(D) 10

2. EASY · NO CALCULATOR NEEDED

$$y = 3x - 5$$

$$6x - 2y = 10$$

How many solutions does the given system of equations have?

(A) Exactly two

(B) Zero

(C) Infinitely many

(D) Exactly one

3. EASY · NO CALCULATOR NEEDED

$$4x + y = 17$$

$$x + 4y = 13$$

The system of equations above is satisfied by (x, y) . What is the value of $x + y$?

4.

EASY · NO CALCULATOR NEEDED

At a school play, adult tickets cost \$8 each and student tickets cost \$5 each. A total of 120 tickets were sold for \$735. If a is the number of adult tickets and s is the number of student tickets sold, which system of equations represents this situation?

(A) $a + s = 120$ and $8a + 5s = 735$

(B) $a + s = 735$ and $8a + 5s = 120$

(C) $a + s = 120$ and $5a + 8s = 735$

(D) $a + s = 120$ and $8a - 5s = 735$

5.

EASY · NO CALCULATOR NEEDED

$$2x + y = 11$$

$$kx - y = 4$$

In the system of equations above, k is a constant. If $(3, 5)$ is the solution to the system, what is the value of k ?

(A) $-\frac{1}{3}$

(B) 1

(C) $\frac{4}{3}$

(D) 3

6.

MEDIUM · NO CALCULATOR NEEDED

$$3x - 4y = 10$$

One equation of a system of two linear equations is shown. The system has no solution. Which of the following could be the second equation?

(A) $3x + 4y = 10$

(B) $6x - 8y = 20$

(C) $6x - 8y = 15$

(D) $8x - 6y = 20$

7.

MEDIUM · NO CALCULATOR NEEDED

Which of the following systems of linear equations has infinitely many solutions?

(A) $2x - 5y = 10$

(B) $4x + 6y = 8$

$6x - 15y = 20$

$6x + 9y = 12$

(C) $x - 3y = 4$

(D) $3x + y = 7$

$2x - 6y = 9$

$6x + 3y = 14$

8.

MEDIUM · NO CALCULATOR NEEDED

$x - 4y = 7$

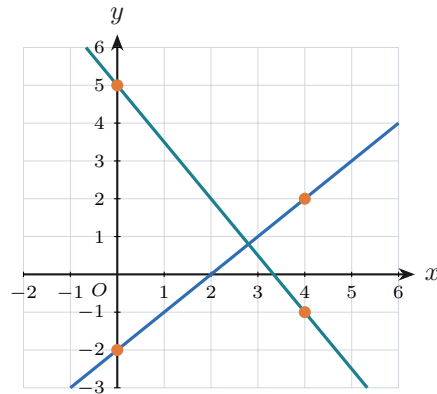
$3x + 2y = 7$

If (x, y) is the solution to the system of equations above, what is the value of $4x - 2y$?

9.

MEDIUM · DESMOS

The graph of a system of two linear equations in the xy -plane is shown. The system has exactly one solution (x, y) . What is the value of y ?



- (A) $\frac{7}{5}$
(C) $\frac{16}{5}$

- (B) $\frac{14}{5}$
(D) $\frac{4}{5}$

10.

MEDIUM · NO CALCULATOR NEEDED

$$2x + ky = 8$$

$$6x + 9y = 20$$

In the system of equations above, k is a constant. If the system has no solution, what is the value of k ?

11.

MEDIUM · NO CALCULATOR NEEDED

A collection of 84 stamps contains only domestic, regional and international stamps. There are twice as many regional stamps as domestic stamps, and there are 6 fewer international stamps than regional stamps. How many international stamps are in the collection?

(A) 30

(B) 36

(C) 18

(D) 24

12.

MEDIUM · CALCULATOR

A coffee shop sold 64 cups of coffee one morning. Small cups cost \$3 each and large cups cost \$5 each, and the total sales were \$270. How many large cups were sold?

13.

MEDIUM · CALCULATOR

Leila drives from home to a lake at an average speed of 60 miles per hour and returns by the same road at an average speed of 40 miles per hour. Her total driving time for the round trip is 5 hours. How many miles is the lake from her home?

14.

HARD · NO CALCULATOR NEEDED

$$6x + 9y = 21$$

One equation of a system of two linear equations is shown. The system has exactly one solution. Which of the following could be the second equation?

(A) $2x + 3y = 11$

(B) $9x + 6y = 21$

(C) $-4x - 6y = -14$

(D) $y = -\frac{2}{3}x + \frac{7}{3}$

15.

HARD · NO CALCULATOR NEEDED

$$ax + by = 13$$

$$bx + ay = 12$$

In the system above, a and b are constants. If $(3, 2)$ is the solution to the system, what is the value of $a^2 - b^2$?

16.

HARD · NO CALCULATOR NEEDED

$$3x + y = 7$$

$$x + 3y = 9$$

If (x, y) is the solution to the system of equations above, what is the value of $5x - y$?

17.

HARD · NO CALCULATOR NEEDED

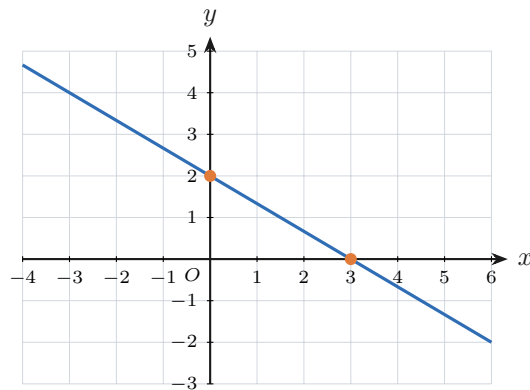
For each real number t , line ℓ_t in the xy -plane is defined by $y = (t + 3)x - 2t + 1$. Which of the following points lies on ℓ_t for every value of t ?

(A) $(7, 2)$ (B) $(1, 4)$ (C) $(2, 7)$ (D) $(2, 5)$

18.

HARD · DESMOS

Line ℓ is shown in the xy -plane. The system consisting of line ℓ and the line $kx + 6y = 5$, where k is a constant, has exactly one solution. Which of the following could NOT be the value of k ?



(A) 4

(B) 5

(C) -2

(D) 1

19.

HARD · CALCULATOR

Priya practices by cycling at a constant speed of 15 miles per hour and by running at a constant speed of 6 miles per hour. In one session she covers a total of 27 miles in 2 hours and 30 minutes. How many minutes did she spend running?

20.

HARD · CALCULATOR

A metalworker melts together an alloy that is 20% silver and an alloy that is 50% silver to make 12 kilograms of a new alloy that is 26% silver. How many kilograms of the 20% silver alloy are used?

(A) 2.4 (B) 3.12

(C) 6 (D) 9.6

21.

HARD · DESMOS

The table shows values of two linear functions, f and g . For which value of x is $f(x) = g(x)$?

x	$f(x)$	$g(x)$
0	5	20
2	11	16
4	17	12

(A) 14 (B) 2

(C) 3 (D) 4

22.

HARD · NO CALCULATOR NEEDED

$$3x + 2y = 10$$

$$6x + ky = m$$

In the system above, k and m are constants. Consider the following statements.

I. If $k = 4$ and $m = 20$, the system has infinitely many solutions.

II. If $k = 4$ and $m = 18$, the system has no solution.

III. If $k = 3$ and $m = 20$, the system has no solution.

Which of the statements are true?

(A) I only

(B) II only

(C) I and II only

(D) I, II, and III

23.

HARD · NO CALCULATOR NEEDED

$$\frac{1}{2}x + \frac{1}{3}y = 7$$

$$\frac{1}{3}x + \frac{1}{2}y = 8$$

If (x, y) is the solution to the system of equations above, what is the value of $x + y$?

24.

SAT CHALLENGE · DESMOS

For each real number t , line ℓ_t in the xy -plane is defined by $(t + 2)x + (1 - t)y = 4t + 5$. All of these lines pass through the same point P . Line m has equation $y = kx - 5$, where k is a constant, and passes through P . What is the value of k ?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$kx + 4y = 8$$

$$9x + ky = 12$$

In the system of equations above, k is a constant. If the system has no solution, what is the value of k ?

(A) -9 (B) -6 (C) 6 (D) 12

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$2x + y = k + 4$$

$$x - y = 2k - 1$$

In the system of equations above, k is a constant. The solution (x, y) to the system satisfies $x = 2y$. What is the value of k ?

(A) 2

(B) 5

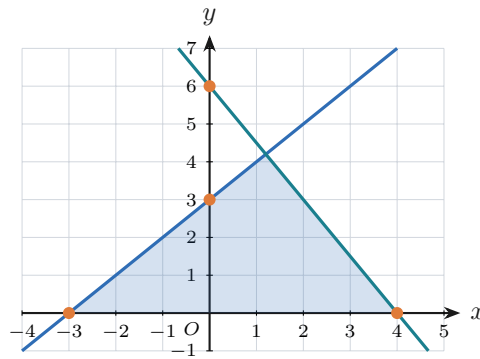
(C) 0

(D) 1

27.

SAT CHALLENGE · DESMOS

The graph shows two lines in the xy -plane. The two lines and the x -axis form a triangle (shaded). What is the area of the triangle?



28.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$6x - 9y = 15$$

$$2x + by = c$$

In the system of equations above, b and c are integers and $1 \leq c \leq 10$. For how many ordered pairs (b, c) does the system have no solution?

29.

SAT CHALLENGE · CALCULATOR

Marcus jogs for h hours at 8 miles per hour. He then walks for twice as many hours as he jogged, at 3 miles per hour, and finally bikes at 12 miles per hour for 30 minutes less than the time he jogged. The total distance of the trip is 46 miles. How many minutes did the whole trip take?

(A) 450

(B) 480

(C) 7.5

(D) 420

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

A box contains only red, green and blue marbles. The ratio of red marbles to green marbles is 2 to 3, and the ratio of green marbles to blue marbles is 4 to 5. After 9 red marbles are removed and 9 blue marbles are added, there are exactly three times as many blue marbles as red marbles. How many marbles were in the box originally?

31.

SAT CHALLENGE · CALCULATOR

A science department orders microscopes at \$240 each and slide kits at \$15 each. The order must include at least 6 slide kits for every microscope, the total cost may not exceed \$2,500, and the storage cabinet holds at most 46 items (microscopes and kits combined). What is the greatest number of microscopes the department can order?

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + 2y = 12$$

$$9x + 6y = 36$$

For the system above, which of the following ordered pairs (x, y) , where t is any real number, is a solution to the system?

(A) $(4 - 2t, -3t)$

(B) $(2t + 4, -3t)$

(C) $(2t + 4, 3t)$

(D) $(3t + 4, -2t)$

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$6x - 2(y - 5) = 2x + 3y + 1$$

$$ax + 20y = b$$

In the system above, a and b are constants. If the system has infinitely many solutions, what is the value of $\frac{b}{a}$?

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$ax + ay = 15$$

$$3ax - 2ay = 20$$

In the system of equations above, a is a nonzero constant. The ordered pair $(2, y)$ is a solution to the system. What is the value of a ?

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{x}{6} + \frac{y}{4} = 2$$

Which of the following equations, when paired with the equation above, forms a system of two linear equations in x and y that has no solution?

(A) $4x - 6y = 40$

(B) $4x + 6y = 48$

(C) $4x + 6y = 40$

(D) $6x + 4y = 48$

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + ky = 12$$

$$21x + 14y = 6m$$

In the system of equations above, k and m are constants. If the system has infinitely many solutions (x, y) , what is the value of km ?